Please upload your completed assignment AND any R Code used to complete the work on Moodle.

---

1. **For this question you will use the FIFA values and FIFA wages datasets. Briefly, these two datasets contain metrics about football (or for me, soccer) players that are represented in FIFA 21.**
   The values dataset contains the following for each player: a unique ID, name, nationality, age, club, all positions played, height, weight, preferred foot, best position, an aggression score, their value, and a multiplier for their value.
   The wages dataset contains the following for each player: a unique ID, name, club, their wage, and a multiplier for their wage.
   Across both datasets, a multiplier of M means 1000000 (one million), a multiplier of K means 100000, and a multiplier of 0 means 0.

   a. **Import both datasets in R**
   b. **For the values dataset create 3 new columns using tidyverse functions:**
      - **One column that contains each player's value multiplier in numeric format (so if a player has a multiplier of M, this new column will have 1000000, and so on for K and 0)**
      - **One column that contains each player's value as a number. This will be the value column times the column you created above**
      - **One column that categorizes player's aggressiveness using the following guide:**
        An aggression score between 0-20 should be labeled "lowest",
        An aggression score between 20-40 should be labeled "low",
        An aggression score between 40-60 should be labeled "neutral",
        An aggression score between 60-75 should be labeled "high",
        An aggression score >= 75 should be labeled "highest"
   c. **For the wages dataset create 2 new columns using tidyverse functions:**
      - **One column that contains each player's wage multiplier in numeric format (so if a player has a multiplier of M, this new column will have 1000000, and so on for K and 0)**
      - **One column that contains each player's wage as a number. This will be the wage column times the column you created above**
   d. **Merge the two datasets into one new dataset**
   e. **Using tidyverse functions, what are the mean wages for each aggression category you created in part b? Are players that are the most aggressive paid the most?**

- **Lowest:** 3,767

- **Low:** 5,270
- **Neutral:** 5,969
- **High:** 9,323
- **Highest:** 23,187

  Players in the "highest" aggression category have the highest mean wage of (23,187).

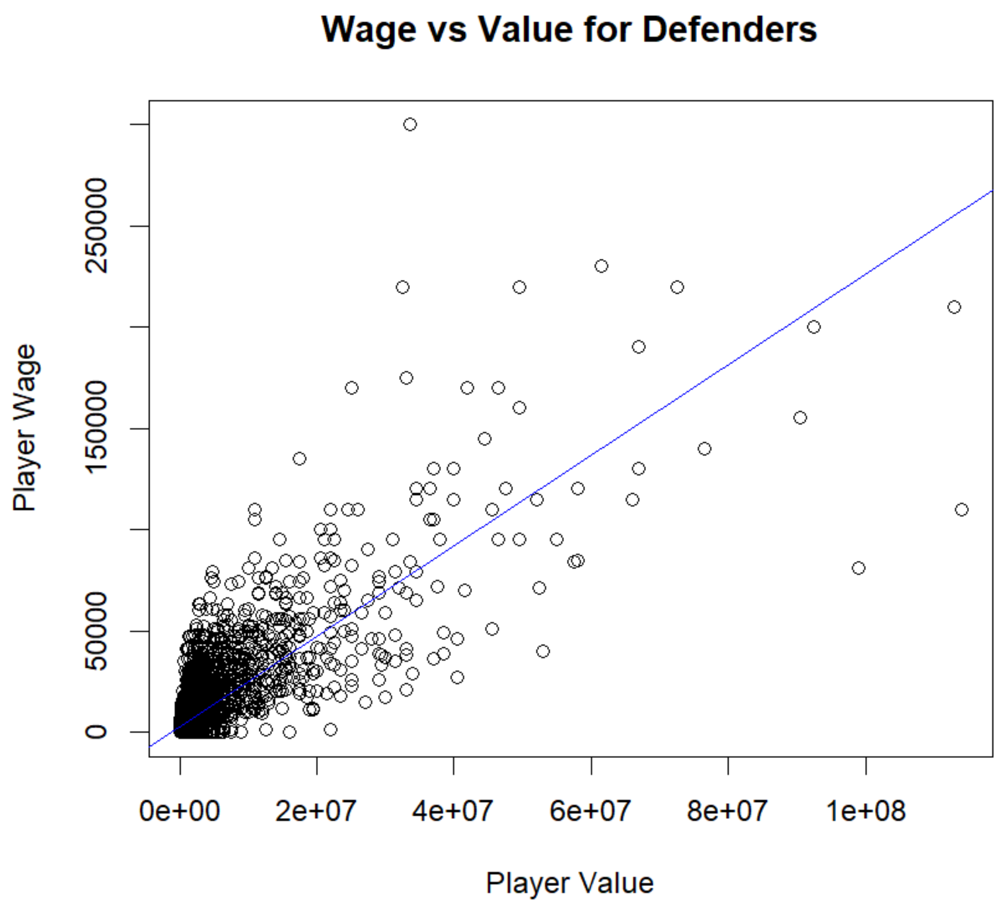
  This suggests that the most aggressive players are paid the most on average, as their mean wage is substantially higher than all other categories.

f. **If you wanted to conduct a hypothesis test to see if the average wage differs among the 5 aggressiveness categories, what analysis technique would you use? (You may need to look at your DATAX111/DATAX121 notes)**

   The appropriate analysis technique is a One-Way ANOVA, since we are comparing the mean wage across more than two groups (the five aggression categories).

g. **Use the filter function to create a new data.frame object that contains only players with best positions of "CB", "LB", and "RB".**

h. **Using the data.frame created in part g, create a scatterplot that shows how a player's wage changes with respect to a player's value. Make sure you have axis labels and that you are plotting the independent variable (wage) on the y-axis and the dependent variable (value) on the x-axis.**


Wage vs Value for Defenders
Player Wage
Player Value
0
50000
150000
250000
0e+00
2e+07
4e+07
6e+07
8e+07
1e+08